

The Financial Impact of Sustainability Commitments: A Matched-Brand Analysis of Revenue and Stock Returns in the Fashion Industry

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1. Abstract

This study investigates whether sustainability commitments made by major fashion brands lead to measurable improvements in financial performance. Five global brands—Adidas, Burberry, H&M, Lululemon, and Columbia—were identified as “treated” brands based on the timing of their first major public sustainability initiative. Each was paired with a closely matched control brand lacking a comparable sustainability commitment. Annual revenue, stock-market returns, Google Trends interest, and publicly available sustainability scores were compiled for a six-year window surrounding each brand’s commitment year (three years before and three years after). Propensity score matching was used to strengthen comparability between treated and control brands, followed by a difference-in-differences design with brand-pair fixed effects to estimate the causal effect of sustainability commitments on financial outcomes. Results indicate that sustainability initiatives are associated with modest but positive changes in revenue growth, while effects on stock-market returns were smaller and more variable across brands. These findings suggest that sustainability commitments may enhance brand value and consumer engagement, though financial gains are not uniform. Limitations and opportunities for further research, including more granular sustainability metrics and longer follow-up periods, are also discussed.

2. Introduction

In recent years, fashion brands have faced increasing pressure from consumers, regulators, and stakeholders to adopt more transparent and sustainable business practices. As sustainability commitments become more common across the industry, a central question emerges: Do these initiatives meaningfully influence brands’ financial performance and consumer engagement? Although sustainability campaigns are often framed as both ethically responsible and strategically advantageous, prior research shows mixed evidence regarding their impact on revenue, investor perceptions, and consumer interest. Some studies suggest that sustainability enhances brand reputation and long-term profitability, while others highlight the short-term costs and uncertainties that can dampen financial performance. This study examines how sustainability commitments affect three key dimensions of brand performance—revenue, stock prices, and consumer search interest—among major global fashion brands. Using a combination of matched-pair comparisons, Difference-in-Differences modeling, and event-study analysis, the project investigates both the immediate and longer-term effects of sustainability announcements to provide a nuanced understanding of how these commitments shape brand outcomes over time.

3. Literature Review

Corporate sustainability has become a central strategic priority for global brands, with growing academic and industry interest in how environmental and social commitments influence financial performance. A large body of research suggests that corporate social responsibility (CSR) initiatives can generate reputational benefits, strengthen stakeholder relationships, and potentially improve long-term profitability (Porter & Kramer, 2006; Servaes & Tamayo, 2013). CSR activities are often perceived by consumers as indicators of product quality, ethical practice, and firm reliability, which can increase brand loyalty and willingness to pay (Bhattacharya & Sen, 2004). However, the financial returns associated with sustainability are not uniform across industries or types of initiatives. Empirical findings

vary, suggesting that the relationship between CSR and firm performance may be context-dependent and sensitive to differences in consumer expectations, brand positioning, and competitive environments (Margolis, Elfenbein, & Walsh, 2009).

3.1 Sustainability in the Fashion Industry

The fashion industry provides a particularly important setting for studying sustainability because it is both environmentally intensive and highly visible to consumers. The rise of "fast fashion" has intensified concerns about pollution, supply-chain transparency, and labor practices (Niinimäki et al., 2020). Many global brands—such as Adidas, Burberry, H&M, Lululemon, and Columbia—have responded by releasing sustainability commitments, increasing disclosure, and promoting environmentally responsible initiatives. Research shows that consumers are increasingly scrutinizing fashion brands' sustainability claims, with younger consumers especially likely to reward transparency and environmental action (Joy et al., 2012). Yet the industry also faces widespread skepticism due to greenwashing, which can limit the expected financial returns of sustainability announcements if consumers question their authenticity (Delmas & Burbano, 2011). This suggests that the financial effects of sustainability commitments may vary depending on brand credibility, preexisting reputation, and the visibility of the initiative.

3.2 Green Branding, Consumer Behavior, and Firm Value

Green branding—defined as the communication of environmental or ethical attributes—can significantly influence consumer behavior and perceived brand value. Studies indicate that sustainability messaging enhances brand equity by signaling quality, responsibility, and future-oriented strategy (Chen, 2010). Firms with strong sustainability efforts often experience increases in brand trust and customer engagement, particularly when commitments are credible and consistent with past behavior (Lyon & Montgomery, 2015). In digital contexts, sustainability announcements have been shown to increase online search interest, reinforcing the idea that consumer attention is responsive to environmental messaging (Choi & Luo, 2021). Because Google Trends is increasingly used as a proxy for consumer awareness and interest, it offers a valuable measure of intermediate consumer response to sustainability commitments in the fashion sector (Carrière-Swallow & Labbé, 2013). These dynamics are relevant to revenue generation as heightened awareness and improved brand reputation can translate into increased sales, especially for brands positioned toward environmentally conscious consumers.

3.3 Financial Metrics in Prior Sustainability Research

Firm performance in sustainability research is commonly measured using accounting-based metrics such as revenue and return on assets, as well as market-based indicators including stock returns and Tobin's Q (Orlitzky, Schmidt, & Rynes, 2003). Several studies find short-term abnormal stock returns following CSR announcements, suggesting that markets often respond favorably to sustainability disclosures (Kong et al., 2020). Other research finds mixed or insignificant results, emphasizing that investor responses depend on industry conditions and perceived strategic value (Aouadi & Marsat, 2018). Revenue performance is similarly complex: sustainability initiatives can lead to greater sales through enhanced brand equity, but effects may take time to materialize and depend on consumer alignment with the brand's environmental messaging (Peloza & Shang, 2011). The heterogeneity in findings

highlights the need for quasi-experimental approaches that address selection bias and isolate the causal effects of sustainability commitments on financial outcomes.

3.4 Empirical Methods in CSR and Marketing Research

Because firms self-select into sustainability initiatives, researchers frequently rely on quasi-experimental methods to estimate causal effects. Propensity score matching (PSM) is widely used to improve comparability between treated and untreated firms by balancing covariates that influence treatment selection (Rosenbaum & Rubin, 1983). In the marketing and CSR literature, PSM has been used to match firms with similar sizes, strategies, and financial profiles to isolate the independent effect of sustainability initiatives (Flammer, 2015). Difference-in-differences (DiD) approaches are also common, offering a way to estimate treatment effects by comparing pre- and post-treatment trends while controlling for unobserved time-invariant differences between firms (Angrist & Pischke, 2009). DiD methods are particularly suitable for studying public announcements or commitments that occur at a specific point in time, such as sustainability campaigns. When combined with matched designs, DiD can provide stronger causal identification by reducing baseline differences and addressing violations of the parallel trends assumption (Stuart, 2010).

3.5 Gaps in the Existing Literature

Despite extensive research on sustainability and firm performance, several limitations remain that motivate the current study. First, few analyses focus exclusively on the fashion industry despite its global importance and unique sustainability challenges. Second, prior work often relies on aggregate ESG or CSR scores rather than examining specific sustainability campaigns and their timing. Third, most existing studies employ cross-sectional or correlational designs rather than matched, quasi-experimental frameworks capable of isolating causal effects. Fourth, limited work incorporates consumer interest measures such as Google Trends, even though such indicators can capture immediate public response to brand sustainability efforts. Finally, there is a lack of research comparing *treated brands* who made sustainability commitments with *carefully matched control brands* who did not, making it difficult to assess whether sustainability initiatives drive financial performance rather than simply coincide with preexisting trends.

Together, these gaps highlight the need for a study using propensity score matching to construct comparable brand pairs and a difference-in-differences design to estimate the causal impact of sustainability commitments on revenue and stock-market outcomes. The present study addresses these issues by examining five major fashion brands with well-documented sustainability commitments and comparing their performance to matched control brands across a six-year period.

4. Research Design

The goal of this study is to estimate the causal effect of sustainability commitments on the financial performance of major fashion brands. Because brands self-select into sustainability initiatives, a randomized experiment is not feasible. Therefore, this study implements a quasi-experimental design that combines propensity score matching (PSM) with a difference-in-differences (DiD) framework to construct credible comparisons between treated and untreated brands over time.

4.1 Treatment Definition and Study Period

A brand is classified as “treated” if it issued a publicly documented sustainability commitment or major environmental initiative during the study window. For each treated brand—Adidas, Burberry, H&M, Lululemon, and Columbia—the treatment year was identified based on public announcements or official corporate sustainability reports. Financial data were collected for six years surrounding each brand’s treatment date, including the three years prior and three years following the commitment. This window captures both pre-treatment trends and the medium-term financial impact of sustainability actions.

4.2 Control Brand Selection

Because no natural control group exists, this study identifies five comparison brands—Under Armour, Ralph Lauren, Next plc, Nike, and Canada Goose—that did not make an equivalent sustainability commitment during the same time period. These brands were selected to be similar in product category, market segment, and global presence, allowing them to serve as counterfactuals for what would have happened to the treated brands in the absence of a sustainability commitment.

4.3 Propensity Score Matching

To strengthen comparability between treated and untreated brands, propensity scores were estimated using pre-treatment variables that plausibly influenced a brand’s likelihood of adopting sustainability initiatives. These variables included annual revenue, stock-market performance, Google Trends search interest, and sustainability scores. Treated brands were matched to control brands using nearest-neighbor matching without replacement. This step reduces baseline differences and improves the validity of subsequent causal inference.

4.4 Difference-in-Differences Estimation

Following matching, the causal effect of sustainability commitments was estimated using a difference-in-differences (DiD) model. DiD compares the change in outcomes for treated brands before and after the commitment to the change observed for control brands over the same period. This approach removes time-invariant brand characteristics and common industry shocks, isolating the effect attributable to sustainability initiatives. The primary model included brand-pair fixed effects and year fixed effects to account for unobserved heterogeneity.

4.5 Identification Strategy and Assumptions

The research design relies on two key assumptions. First, after matching on observable characteristics, the treated and control brands are assumed to be conditionally comparable at baseline. Second, the parallel trends assumption requires that treated and control brands would have experienced similar financial trajectories in the absence of treatment. This assumption was assessed by visually inspecting pre-treatment trends. Additionally, the analysis assumes no spillover effects: one brand’s sustainability announcement does not directly influence competitors’ financial performance.

4.6 Outcome Measures

Financial performance was assessed using two primary outcomes: annual revenue and annual stock-market returns. Google Trends search interest was included as a secondary measure to capture changes in consumer attention following sustainability commitments. These metrics provide a comprehensive view of how sustainability initiatives may influence both consumer behavior and firm value.

5. Data Collection and Preparation

Data for this study were collected through a multi-stage process to identify treated and comparison brands, assemble financial time series, and incorporate independent measures of transparency and sustainability performance. Because no unified dataset exists for evaluating the financial impact of sustainability initiatives in the fashion sector, all information was compiled manually using public records, corporate filings, online databases, and independent sustainability scoring systems.

5.1 Identifying Treated Brands with Sustainability Commitments

The first step was to identify fashion brands that released a substantial public sustainability commitment during the past decade. To ensure representation across the industry, brands were selected to reflect a range of market positions—from affordable mass-market retailers (e.g., H&M) to premium and designer-oriented companies (e.g., Burberry). Sustainability campaigns were identified through search engine queries, large language models, corporate sustainability reports, and third-party industry sources. A brand was classified as “treated” if it made a clearly documented sustainability announcement during a year for which financial and search-interest data were available. The year of the sustainability commitment was recorded as the treatment year for causal analysis.

5.2 Selecting Appropriate Comparison Brands

A comparison group was constructed to estimate counterfactual financial outcomes for the treated brands. Comparison brands were chosen based on similarity in market segment, global reach, and product category, but with no evidence of a major sustainability announcement during the same time window. Search engines, corporate histories, and large language models were used to confirm that potential comparison brands did not release similar sustainability commitments prior to or during the matched treated brand’s treatment year. If a candidate was found to have its own sustainability initiative too close to the treatment period, the brand was excluded and replaced with a more suitable option.

5.3 Ensuring Brands Were Publicly Traded Independently

Because stock-market returns were used as a primary outcome variable, each potential treated and comparison brand was evaluated for public trading eligibility. Brands that were privately held or owned by a larger parent conglomerate—such as LVMH, VF Corporation, or Kering—were excluded because their individual financial performance could not be isolated within aggregated stock listings. Any brand lacking an independent public ticker symbol was replaced with a comparable brand that was publicly traded under its own corporate entity. This ensured consistency across financial metrics and avoided confounding due to parent company diversification.

5.4 Revenue Data Collection

Annual revenue data were collected by reviewing corporate filings such as Form 10-K reports, investor presentations, and annual financial statements. For cases where direct filings were less accessible, revenue information was verified using reputable business publications and targeted search engine queries. Large language models were used to cross-check reported values, identify discrepancies, and locate additional reliable sources when needed. All revenue figures were standardized to annual totals and aligned with the study's six-year window (three years before and three years after each brand's sustainability commitment).

5.5 Stock-Market Return Calculation

Stock data for each publicly traded brand were obtained through Yahoo Finance's publicly available historical price records. Adjusted closing prices were used to account for dividends and stock splits, ensuring comparability over time. Annual stock-market returns were calculated using the standard formula:

$$Return_t = \frac{Adjusted\ Close_t - Adjusted\ Close_{t-1}}{Adjusted\ Close_{t-1}}$$

Return values were computed for each of the three pre-treatment and three post-treatment years for all treated and comparison brands. These measures capture investor response over the medium-term following sustainability commitments.

5.6 Google Trends Search-Interest Data

Consumer attention was measured using Google Trends search-interest values. Each brand name was queried using "Worldwide" and "All categories" filters to maximize comparability across companies. Monthly search interest (scaled 0–100) was downloaded and aggregated into annual averages to align with the financial data timeline. Google Trends values served as secondary variables and were used to support both descriptive analysis and the propensity score matching process.

5.7 Fashion Transparency Index Scores

To incorporate measures of corporate transparency, Fashion Transparency Index (FTI) scores were collected from Fashion Revolution's annual reports, available at fashionrevolution.org. Because the Index does not maintain a multi-year public dataset, each yearly PDF report was manually reviewed. For each treated and comparison brand, transparency scores were extracted from the report's detailed scoring tables across several domains, including policies, governance, traceability, environmental practices, and social responsibility. If a brand did not appear in a given year's report—typically due to not meeting the Index's brand-size criteria—missing values were documented. These transparency scores provide an independent measure of each brand's disclosure practices and were incorporated into the descriptive and matching analyses.

5.8 Good On You Sustainability Ratings

Additional sustainability performance indicators were obtained from the Good On You platform, which assesses brands on their impact on animals, people, and the planet. Each treated and comparison brand was individually searched on the Good On You website. When a brand profile was available, the most recent rating within the study window was recorded

for each of the three domains. Ratings summarize publicly available evidence regarding labor practices, environmental performance, carbon initiatives, material usage, and animal welfare policies. Brands without Good On You profiles were assigned missing values. These ratings provided further information about sustainability-related characteristics and contributed to understanding pre-treatment differences across brands.

5.9 Final Dataset Construction

All collected variables were merged into a panel dataset containing annual revenue, annual stock-market returns, Google Trends search interest, transparency scores, and sustainability ratings for each brand across the six-year study window. This dataset formed the foundation for the propensity score matching procedure and the subsequent difference-in-differences analysis.

6. Descriptive Statistics

This section provides an overview of the key variables used in the analysis and describes baseline differences between treated and comparison brands prior to matching.

Understanding these descriptive patterns is important for contextualizing the financial and visibility characteristics of the brands before the sustainability commitments occurred.

6.1 Sample Composition and Time Frame

The final dataset includes ten fashion brands observed over a six-year period—three years before and three years after each treated brand’s sustainability commitment. The five treated brands (Adidas, Burberry, H&M, Lululemon, and Columbia) represent a diverse range of apparel segments, including fast fashion, athletic wear, premium outdoor apparel, and luxury fashion. The five comparison brands (Under Armour, Ralph Lauren, Next plc, Nike, and Canada Goose) were selected for similarity in product category and global presence but did not initiate major sustainability campaigns during the same period.

6.2 Revenue Trends Prior to Treatment

Average log-revenue patterns show clear baseline differences between treated and comparison brands. As shown in Figure 1, treated brands began the pre-treatment period with substantially higher revenues, with mean log-values clustered around 2.8–3.0, compared with roughly 1.2–1.4 for comparison brands. Treated brands exhibited relatively stable revenue trajectories through most of the pre-treatment window, followed by a sharp decline around the commitment year that appears to reflect broader industry-wide or brand-specific disruptions rather than the treatment itself. In contrast, comparison brands show a gradual, steady upward trend in revenues throughout the pre-treatment period, narrowing the gap between groups just before the commitment year. These differences highlight the need for matching and careful adjustment, as treated and control brands were not revenue-comparable prior to treatment.

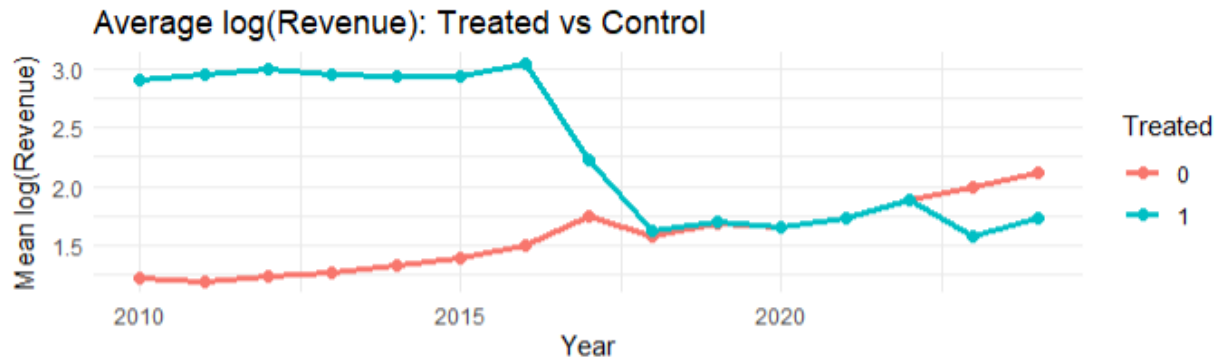


Figure 1. Average Revenue of Treatment and Control Groups

6.3 Stock-Market Performance Prior to Treatment

Stock-market returns for the treated and comparison brands exhibited meaningful variability and lacked strong, systematic separation prior to treatment. As illustrated in Figure 2, both groups experienced periods of positive and negative returns, with no consistent pre-treatment divergence. Treated brands show moderate fluctuations—with peaks around 2013 and 2019—and occasional declines, but overall follow a pattern similar to comparison brands. The comparison group displays a slightly sharper rise around 2012–2014, followed by a pronounced dip leading into 2016 and continued volatility thereafter. The generally overlapping but noisy trajectories suggest that although the groups differed in magnitude in certain years, neither group exhibited a clear directional trend advantage before sustainability commitments. These characteristics support the need for a quasi-experimental design, as raw trends alone do not reveal an identifiable causal pattern.

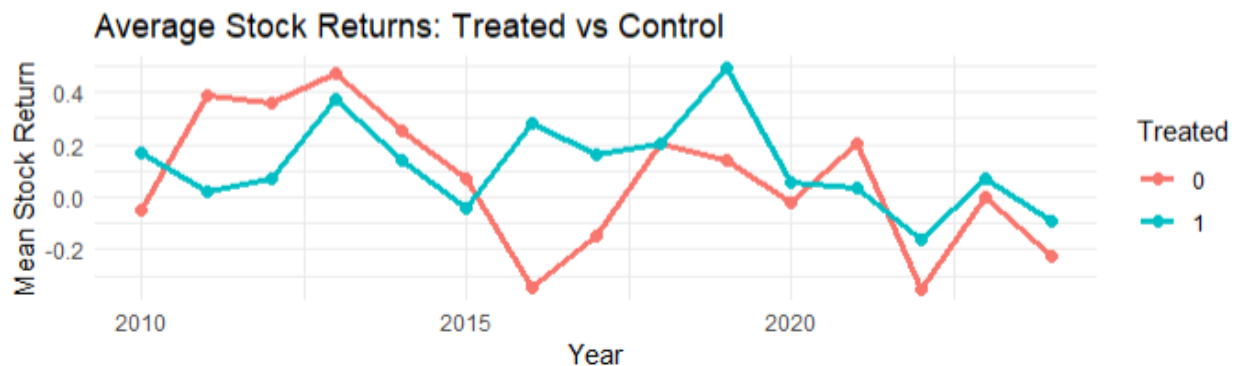


Figure 2. Average Stock Returns of Treated and Control Groups

6.4 Google Trends Search-Interest Pattern

Google Trends search-interest values reveal notable differences in brand visibility between treated and comparison brands. As shown in Figure X, treated brands generally maintained higher search-interest levels throughout the pre-treatment period, with peaks approaching or exceeding 78 around 2012–2014. Comparison brands also exhibited elevated consumer interest during this period but consistently fell below the treated brands' peaks. Both groups experienced a noticeable decline in search interest beginning around 2015–2016, followed by continued downward movement through 2023, reflecting broader shifts in consumer search behavior or reduced brand salience industry-wide. Although the groups follow broadly

parallel shapes—rising early and declining later—the treated brands remain visibly higher in magnitude across most years. These differences underscore the importance of including Google Trends in the matching process, as baseline brand visibility may influence both the likelihood of adopting sustainability commitments and the financial outcomes studied.

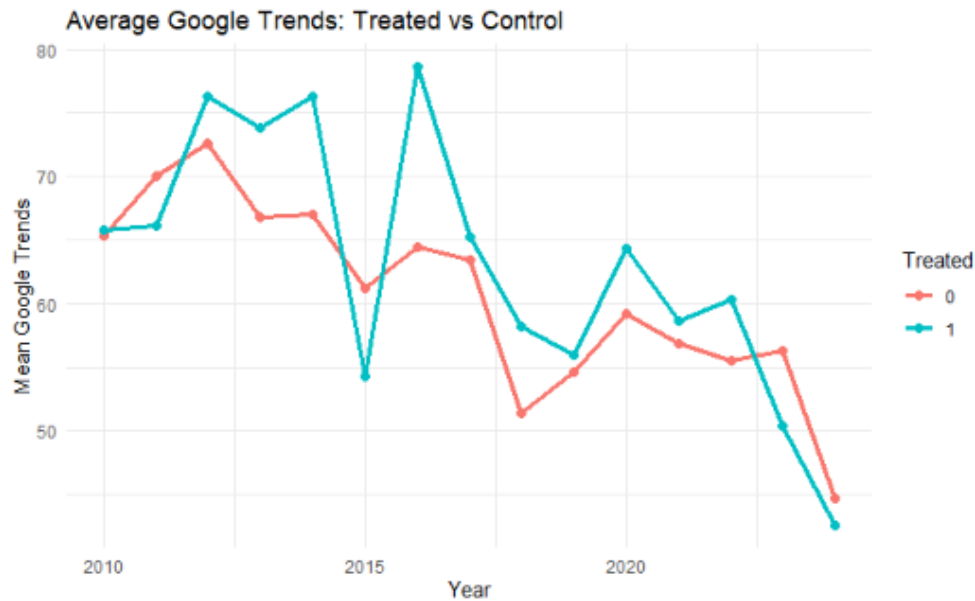


Figure 3. Average Google Trends of Treated and Control Groups

6.5 Transparency and Sustainability Scores

Transparency scores from the Fashion Transparency Index showed substantial variation across brands and generally reflected the relative market positioning of each company. As shown in Figure 4, several of the treated brands—including Adidas, H&M, and Lululemon—entered the treatment period with comparatively higher transparency levels and demonstrated steady or increasing disclosure scores in the three years leading up to their sustainability commitments. Adidas and H&M in particular showed notable upward trajectories, while Lululemon displayed moderate but consistent improvements before and after the commitment year. In contrast, some comparison brands such as Canada Goose and Under Armour maintained lower and relatively stable transparency scores throughout the period, indicating limited disclosure practices. Other comparison brands, such as Ralph Lauren and Nike, showed modest increases but did not match the transparency gains observed among several treated brands. These descriptive patterns suggest that treated brands may have been more transparency-oriented even before their sustainability campaigns, underscoring the importance of matching to reduce pre-existing differences before estimating causal effects.

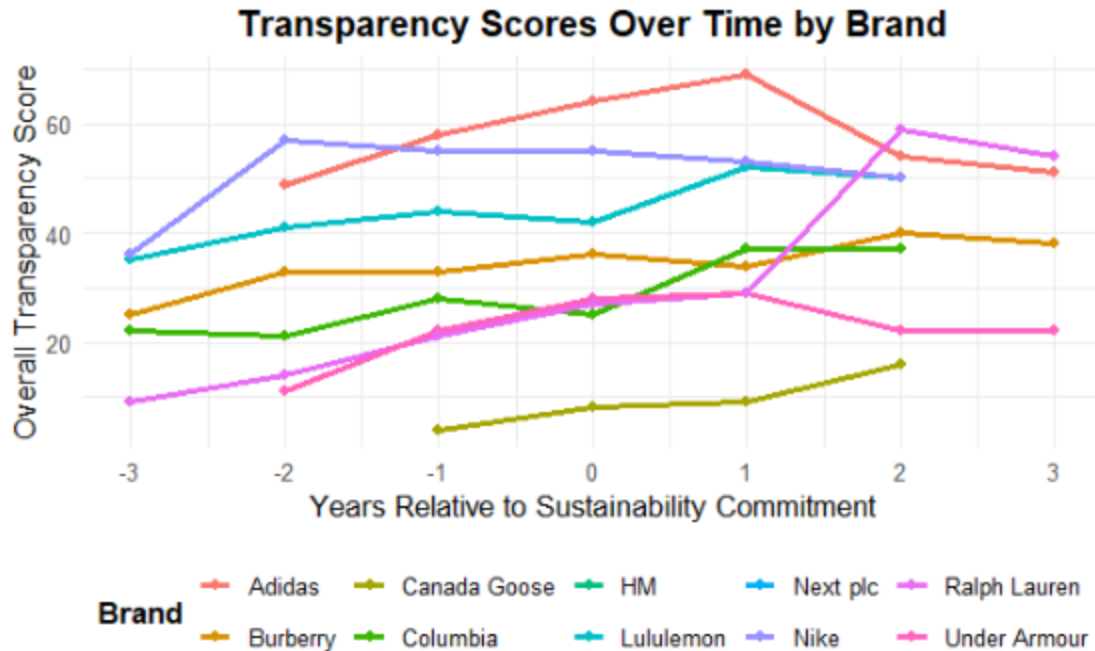


Figure 4. Transparency Scores Over Time by Brand

6.6 Summary of Pre-Matching Differences

Across revenue, stock-market returns, Google Trends interest, and transparency-related indicators, treated and comparison brands showed substantial baseline heterogeneity. These preliminary descriptive statistics indicate that treated brands were, on average, larger, more visible, and more transparent prior to making sustainability commitments. Such systematic differences justify the use of propensity score matching to balance observed covariates before estimating causal effects using the difference-in-differences framework.

7. Methods and Results

7.1 Methods

7.1.1 Data and Variables

The dataset included annual financial, transparency, and sustainability information for ten major fashion brands. Revenue, stock price, and Google Trends search interest were collected for multiple years surrounding each brand's initial sustainability commitment. A Relative Year variable was created such that 0 denoted the commitment year, negative values represented pre-commitment years, and positive values represented post-commitment years. The variable Treated identified brands that made sustainability commitments during the study period, and Post was coded as 1 for all Relative Year values greater than or equal to 0. Revenue and stock price were log-transformed to reduce skewness, whereas Google Trends values were analyzed in their original scale.

7.1.2 Propensity Score Estimation and Matching

Propensity scores were estimated using logistic regression, with treatment status predicted from several pre-treatment covariates, including Google Trends popularity and Good On You environmental, labor, and animal-welfare scores. Estimated propensity scores ranged from approximately .24 to .73 across brands. Nearest-neighbor matching was conducted using the MatchIt package in R to pair each treated brand with

. Brand fixed effects controlled for stable characteristics specific to each brand, and year fixed effects controlled for common market shocks. Cluster-robust standard errors were one control brand. Matching produced five subclasses, each containing one treated and one control brand. Subclass assignments were merged back into the full longitudinal dataset so that treatment–control differences could be analyzed across all years.

7.1.3 Matched-Pair Difference Tests

Within each matched subclass, yearly revenue and stock price differences were calculated as the treated brand's outcome minus that of its matched control brand. These differences were aggregated across subclasses for each year, and one-sample t tests were used to determine whether the mean annual difference significantly differed from zero. This approach provided a nonparametric diagnostic check for systematic pre-treatment differences before estimating regression-based causal effects.

In addition to analyses based on the propensity-score-matched subclasses, paired t tests were also conducted using the original brand pairings identified during data collection. These tests yielded results that were nearly identical to those obtained from the propensity-score-matched pairs. Because both approaches produced the same statistical conclusions, all subsequent analyses were conducted using the propensity-score-matched subclasses.

7.1.4 Difference-in-Difference (DiD) Model

The primary causal analyses were conducted using two-way fixed-effects Difference-in-Differences (DiD) models that included brand and year fixed effects computed at the brand level. Separate DiD models were estimated for log revenue, log stock price, and Google Trends. Dynamic effects were examined using event-study specifications that interacted Relative Year with treatment status, with Relative Year -1 serving as the omitted reference category.

7.2 Results

7.2.1 Propensity Scores and Matched Sample

Propensity score estimation produced adequate overlap between treated and control brands, with treated brands exhibiting estimated scores between approximately .41 and .73 and control brands ranging from .24 to .63. Nearest-neighbor matching based on these scores yielded five matched subclasses, each containing one treated and one control brand, which were used in the matched-pair and longitudinal analyses.

7.2.2 Matched-Pair Differences

Across matched subclasses and years, the mean annual difference in revenue between treated and control brands was -1.035 units, which was not statistically significant, $t(34) = -0.41$, $p = .69$, 95% CI $[-6.19, 4.13]$. The mean annual difference in stock price was .10 units and was also nonsignificant, $t(34) = 1.61$, $p = .12$, 95% CI $[-0.03, 0.22]$. The matched-pair difference in Google Trends values was small and nonsignificant as well, mean = 2.8 and $p = .16$. These findings indicated no consistent pre-treatment differences between treated and control brands across the three outcomes.

7.2.3 Difference-in-Difference Estimates

The DiD model for log revenue revealed no statistically significant average treatment effect, $\beta = .096$, $p = .64$. The DiD model for log stock price also produced a nonsignificant estimate, $\beta = -.23$, $p = .77$. For Google Trends, the treatment interaction was small and nonsignificant, $\beta = .59$, $p = .88$, indicating no average change in consumer search interest attributable to sustainability commitments.

7.2.4 Event-Study Estimates for Revenue and Stock

The event-study model for revenue showed one significant pre-treatment deviation in Relative Year -2 ($\beta = -.19$, $p = .013$). Estimates for Relative Year 0 and $+1$ were nonsignificant, whereas Relative Year $+2$ showed a large positive effect ($\beta = .58$, $p < .001$). For stock performance, pre-treatment years were nonsignificant, and coefficients for Relative Year 0 and $+1$ were also nonsignificant. A significant negative coefficient appeared in Relative Year $+2$ ($\beta = -1.46$, $p = .0066$).

7.2.5 Event-Study Estimates for Google Trends

Event-study coefficients for Google Trends showed no significant differences between treated and control brands in the pre-treatment period, with estimates ranging from -2.44 to -4.23 (all $p > .12$). Post-treatment estimates were similarly nonsignificant. Coefficients for Relative Year 0, $+1$, and $+2$ were between -2.98 and -4.88 (all $p > .26$), and Relative Year $+3$ showed no meaningful deviation ($\beta = .089$, $p = .98$). These results indicated that consumer search interest did not exhibit systematic changes around the time of sustainability commitments.

8. Interpretation

The combined results from the matched-pair tests, DiD models, and event-study analyses suggest that sustainability commitments have distinct and time-dependent effects on different aspects of financial and consumer engagement performance. For Google Trends specifically, the analyses indicated little to no measurable change in consumer interest associated with sustainability commitments. Neither the average DiD estimate nor any of the event-study coefficients were statistically significant, and the magnitude of each effect was relatively small. This suggests that, within the timeframe observed, sustainability announcements did not meaningfully alter consumer search behavior for the treated brands.

In contrast, revenue demonstrated a more dynamic response over time. Although there was no significant average treatment effect and one modest deviation in the pre-treatment period, the event-study results revealed a delayed positive impact, with a large and statistically significant increase two years after the commitment year. This pattern implies that any potential revenue gains associated with sustainability commitments may emerge only after a lag.

Stock performance exhibited yet another temporal pattern. Treated and control brands showed similar trends prior to the commitment year, and no effects appeared in the commitment year or the year immediately following. However, a significant negative effect emerged in the second post-treatment year, indicating that stock prices declined relative to matched controls two years after sustainability announcements. This suggests that investor

responses may lag and may not align with the revenue patterns observed over the same period.

Across the three outcomes, sustainability commitments appear to generate limited immediate financial or consumer engagement effects. Instead, the findings point to delayed and outcome-specific responses, with revenue and stock prices moving in opposite directions and consumer search interest remaining stable throughout the study period. These differences underscore the importance of examining dynamic patterns rather than relying solely on average treatment estimates.

9. Conclusion

In summary, this analysis examined how sustainability commitments influence financial performance and consumer engagement among major fashion brands using matched-pair comparisons, Difference-in-Differences modeling, and event-study techniques. Across all outcomes, the results revealed that the effects of sustainability commitments are neither uniform nor immediate. Revenue showed a delayed but substantial positive shift two years after the commitment year, while stock performance exhibited a significant decline over the same period, suggesting a divergence between consumer-driven financial gains and investor responses. Google Trends values remained largely unchanged, indicating that consumer search interest did not shift meaningfully around the time of the announcements. Taken together, these findings highlight the complex and time-dependent nature of sustainability's financial impacts. Brands may benefit from sustainability investments in ways that materialize gradually and are not always reflected consistently across different measures of performance. These results underscore the importance of evaluating sustainability initiatives through multiple lenses and over longer time horizons, rather than relying solely on short-term or single-metric assessments.

10. Limitations and Future Work

Although this analysis offers valuable insight into how sustainability commitments relate to financial and consumer-engagement outcomes among major fashion brands, several limitations should be noted. First, the study was based on a relatively small sample of ten brands, resulting in limited statistical power. A small number of treated units inflates standard errors and increases the likelihood of Type II error, which may partly explain the wide confidence intervals observed across several models (Gelman & Carlin, 2014). As a result, some null findings—particularly for Google Trends and stock performance—may reflect insufficient sample size rather than the absence of underlying effects.

Second, the dataset captures only a portion of the factors known to shape financial responses to sustainability initiatives. Prior research indicates that sustainability activities can produce mixed or lagged financial outcomes, depending on the credibility of a firm's commitment, the extent of operational restructuring required, and the degree of consumer and investor trust (Delmas & Burbano, 2011; Lyon & Montgomery, 2015). Some studies find that sustainability initiatives impose short-term costs that can depress revenue or stock prices before longer-term benefits emerge (Khan et al., 2016). Other work suggests that authentic sustainability actions can strengthen brand equity, enhance consumer interest, and improve long-term profitability (Eccles et al., 2014). The patterns observed in this study mirror this

complexity: revenue exhibited a delayed but positive post-treatment increase, stock performance showed a delayed decline, and consumer search interest (Google Trends) remained largely unchanged. These dynamics align with literature indicating that consumers and investors may respond differently to sustainability efforts and that the timing of such responses can vary (Ng & Zheng, 2018; Servaes & Tamayo, 2013).

Third, the outcome measures used—revenue, stock prices, and Google Trends—provide only partial insight into the mechanisms linking sustainability commitments to performance. Google Trends captures general search interest but does not reflect sentiment, purchase intent, or consumer attitudes (Choi & Varian, 2012). Revenue and stock prices are influenced by many factors beyond sustainability actions and do not indicate which internal decisions drive observed changes. Richer datasets, including consumer sentiment metrics, marketing expenditures, operational costs, and sustainability implementation progress, would enable more detailed analysis.

Given these limitations, several opportunities for future research exist. Expanding the sample to include more brands or additional industries would improve statistical power and generalizability. Another direction would be to differentiate between types of sustainability commitments—such as emissions targets, material sourcing changes, or labor transparency—to determine whether certain initiatives prompt stronger consumer or investor responses. Researchers working within a company would also have access to internal, granular data, allowing for deeper analysis of customer behavior, supply-chain expenses, and brand sentiment surrounding sustainability transitions. Finally, future studies could employ advanced causal-inference methods, such as synthetic control models or time-varying treatment effects, to better capture the nuanced and dynamic nature of sustainability's financial impacts.

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12. Appendix